# Kevin Rivera

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## EDUCATION

**Duke University** *August 2026 – December 2027 (Expected)*
*Master of Engineering in Robotics & Autonomy*
**Relevant Coursework:** Linear System Theory and Machine Learning, Continuum Robotics, Management in High Tech Industries

**University of Central Florida** *May 2022 – May 2026*
*Bachelor of Science, Mechanical Engineering*
**GPA:** 3.51/4.00
**Relevant Coursework:** Intro to Robotics, Vibrations & Controls, Heat Transfer I & II, Thermodynamics, Computer-Aided Engineering Design

## EXPERIENCE

**Agricultural Robotics Lab – UCF** *May 2025 – August 2026*
*Volunteer Undergraduate Research Assistant*

- Developed a novel, low-cost computer vision depth-estimation system for a strawberry-harvesting robot using two consumer RGB webcams and a Raspberry Pi; achieved ~1% error versus measured distance across a 5–25 cm working range, without RGB-D cameras, LiDAR, or neural networks
- Designed the real-time MATLAB perception pipeline (weighted RGB decomposition, binary thresholding, morphological operations, kernel smoothing) that extracts each strawberry's full perimeter from a live webcam feed and isolates its bottom-edge contour for depth estimation
- Designed modular SolidWorks test fixtures and end-effector mounts so a single failed or revised section could be reprinted alone, saving ~6 hours of print time per reprint versus the ~8.5-hour full set
- Primary author of a conference paper in preparation on the method; presented results weekly to the lab director

**Hubbard Construction (Vinci)** *May 2024 – January 2025*
*Estimating Intern*

- Performed quantity takeoffs from engineering plan sets and prepared 30+ asphalt estimates ($200K to multimillion) in Excel and Bluebeam Revu for FDOT, Walt Disney World, Universal, Orlando International Airport, and multiple Florida counties
- My takeoffs supported winning bids including I-4, the Epic Universe bus depot, and Orlando Sanford Airport
- Reconciled takeoff quantities against engineer-predicted quantities and investigated discrepancies, tracing them to mislabeled plans or calculation errors

## PROJECTS

**UCF Senior Design – Bring ROAR to Life** | 5-person team *August 2025 – May 2026*
*Anatomically correct animatronic Stegosaurus, from concept to working prototype*

- Owned electrical design, C++ firmware, and system integration: designed wiring diagrams, then soldered and integrated an ESP32 controlling 11 servos across four 2-DOF legs and a soft continuum tail, plus an IMU, accelerometer, temperature sensor, and audio playback
- Diagnosed false tip-over alarms caused by 12 V servo power noise corrupting low-voltage IMU readings; implemented a 5-sample averaging filter with spike rejection and asymmetric weighting (slow rise, fast decay) that reduced false fail-safe triggers
- Designed safety systems: IMU tip-over and over-temperature detection that lock out operator inputs and alert staff, plus fuses, emergency stops, and fully enclosed electronics for child-safe operation
- Built a long-range teleoperation interface: an ESP32-hosted web app that bridges a PS4 controller through any smartphone, reaching a tested ~100–150 ft range beyond direct-Bluetooth limits
- Streamed live IMU, temperature, controller-input, and terminal telemetry to the phone for real-time debugging during testing
- Made the component and purchasing decisions that delivered the prototype for ~$450 against an $800 budget (~44% under); produced the full cost analysis
- Co-authored multiple 150+ page design reports with stakeholder and risk analysis, including electrical and fire hazards
- Served as point of contact for the Natural History Museum, London and Lawrence Hall of Science (UC Berkeley) on skeletal accuracy; applied gait data to leg motion and, per their feedback, added off-the-shelf parts and hot-swap batteries
- Presented in weekly design reviews to an adjunct professor and full-time NASA engineer; prototype ran 30 minutes to 2 hours per battery and completed 3 demos

**Continuum Robotics – Collapsible SpiRobs Robotic Arm** | ME 555, 4-person team *August 2026 – Present*
*Design, control, and assembly of a collapsible tendon-driven robotic arm*

- Independently iterated the 2D planar arm through 5 SolidWorks design-for-manufacturing revisions, resolving cracking, layer delamination, weak thin connections, and undersized holes by thickening the spine and tuning TPU 95A print settings
- Optimized segment angles and added cutouts so the design collapses ~40% shorter (CAD estimate)

- Extending the design to 3D: diagnosing first-print failures by scaling the model up 30% and revising support settings; ESP32 teleoperation of three nitinol tendons (two for left/right bending, one for extension/contraction) planned

**Observer-Based Fault Detection and Isolation for Industrial Systems** | ME 627, solo *October 2026 – Present*

*Model-based diagnosis of gradual sensor and actuator faults to support predictive maintenance*

- Building a state-observer bank in Python (python-control, NumPy) for residual generation, fault injection, and isolability analysis of sensor drift and loss of actuator authority; completion December 2026

**Wizardry Chess** | 3-person team *January 2026 – May 2026*

*Fully automated chess board that plays live Lichess games via API*

- Assembled and wired the custom XY gantry, stepper motors, and electromagnet actuation controlled by a Raspberry Pi with ROS2
- Validated and reviewed the team's C++ control code; tested in 2 full player-vs-player games at ~2 to 4 seconds per move
- Traced misplaced moves to speech-to-text misrecognition of spoken moves by reading the terminal output of the audio input
- Conceived the "Loyalty Mechanic" (built by a teammate), which triggers insubordinate moves when a player blunders

## LEADERSHIP

**Phi Eta Mu – Delta Chapter Non-Profit** *August 2024 – Present*

*Guardian & Former Canciller, Secretary*

- Led operations and administration of a 40-member chapter as Canciller
- Organized cross-chapter events including Universal Halloween Horror Nights and a year-end camping trip (80+ attendees), raising meeting and event attendance ~20% above typical levels
- Created and sold chapter merchandise, increasing chapter funds by 60%
- Founded and manage the Phi Eta Mu Networking LinkedIn group with supreme council support, growing it to 110+ members in 5 months across all chapters
- Hosted ~7 Zoom panels on careers, personal finance, and LSAT/MCAT prep, each with 3 to 4 panelists and up to 60 attendees
- Awarded the Orden José Menéndez (2026), given to at most 2 members per year since 1981, for service and leadership

**Phi Eta Mu – Committee of Medios Electrónicos** *September 2026 – Present*

*Committee Member*

- Contacted 8 employers to date for the March 2027 career fair (prior fairs: 600 to 800 attendees, 40+ employers)

## CLUBS

**Duke Advanced Degree Consulting Club (DACC)** | Consultant *September 2026 – Present*

- Benchmarked competitors' products, prices, features, geographic reach, and accessory offerings to inform a pro bono client's market positioning (under NDA)

**Duke AERO (Rocketry)** | Member *October 2026 – Present*

- Joining the payload sub-team, working on the rover

## SKILLS

**Software:** MATLAB, Python (NumPy, python-control), C++, ROS2, SolidWorks, Fusion 360, AutoCAD, ANSYS (structural FEA, coursework), Gazebo (coursework), Linux, Git, Microsoft Office, Bluebeam Revu
**Technical:** Computer Vision, Image Processing, Embedded Systems, System Integration, Design for Manufacturing, Wiring Diagrams, Soldering, 3D Printing (TPU), Fault Detection, Quantity Takeoffs, Cost Estimating
**Hardware:** ESP32, Arduino, Raspberry Pi (Pico), Servos, Stepper Motors, IMU, Temperature Sensors
**Languages:** Spanish and English